

IMPERIAL

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

CIVE70100 RESEARCH PROJECT – STRUCTURES 2025–2026

Developing design formulae for flexural buckling of stainless steel columns using machine learning techniques

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1 Introduction

2 Literature review

2.1 Structural stability and the mechanics of non-linear materials

2.1.1 Column buckling and the normalised slenderness approach

Perfect columns are idealised structural members that are perfectly straight and exhibit elastic–perfectly plastic behaviour. The strength of such a member is bounded by two limiting modes of failure: geometric instability and material yielding. The former is governed by the Euler critical buckling load, which for a pin-ended column is

$$N_{\text{cr}} = \frac{\pi^2 EI}{L^2}, \quad (2.1)$$

where E is Young’s modulus, I is the second moment of area and L is the member length (Timoshenko and Gere, 2009). The latter is governed by the squash load $N_y = Af_y$, where A is the cross-sectional area and f_y is the yield stress. Together these define the normalised slenderness $\bar{\lambda}$, which collapses the column’s geometry, material and length into a single dimensionless parameter, while the ultimate load N_u is normalised by the squash load to give the buckling reduction factor χ :

$$\bar{\lambda} = \sqrt{\frac{N_y}{N_{\text{cr}}}}, \quad \chi = \frac{N_u}{N_y}. \quad (2.2)$$

As illustrated in Figure 2.1, the theoretical maximum strength of a perfect column forms a sharp χ – $\bar{\lambda}$ baseline: a pure squashing plateau ($\chi = 1$) for stocky columns and an elastic Euler hyperbola ($\chi = \bar{\lambda}^{-2}$) for slender columns, the two regimes meeting at $\bar{\lambda} = 1$ (Trahair, 2008).

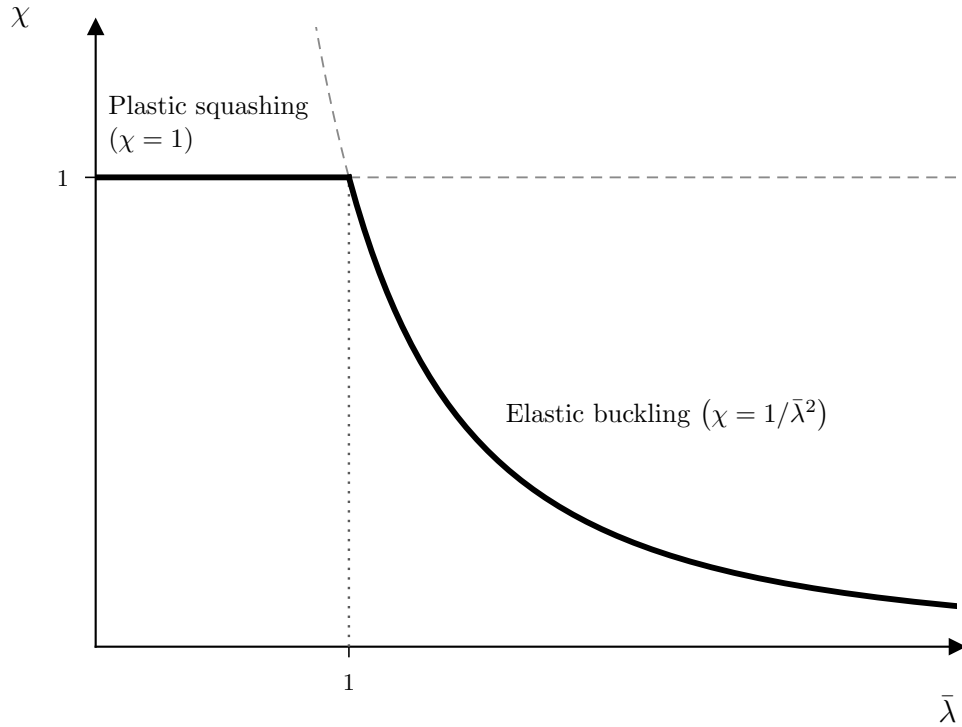


Figure 2.1: Buckling reduction factor χ against normalised slenderness $\bar{\lambda}$ for a perfect, elastic–perfectly plastic column: the lower envelope of the squash plateau ($\chi = 1$) and the Euler hyperbola ($\chi = \bar{\lambda}^{-2}$), which meet at $\bar{\lambda} = 1$.

2.2 Design philosophies for non-linear material buckling

2.2.1 Empirical buckling curves and the Eurocode 3 approach

2.2.2 The tangent-modulus approach

2.3 The Köllner–Gardner–Wadee (2023) energy formulation and its open question

2.3.1 A closed-form solution for Ramberg–Osgood struts

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2.4 The physical drivers of imperfection sensitivity

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Bibliography

- Timoshenko, S. P. and Gere, J. M., eds. (2009). *Theory of Elastic Stability*. 2nd ed., Dover ed. A McGraw-Hill Classic Textbook Reissue. Mineola, N.Y: Dover Publications. 541 pp. ISBN: 978-1-62198-613-3 978-0-486-13480-2.
- Trahair, N. S., ed. (2008). *The Behaviour and Design of Steel Structures to EC3*. Fourth edition. London: Taylor & Francis. 1 p. ISBN: 978-1-315-27351-8. DOI: [10.1201/9781315273518](https://doi.org/10.1201/9781315273518). Available at: <https://www.taylorfrancis.com/books/9781315273518>.